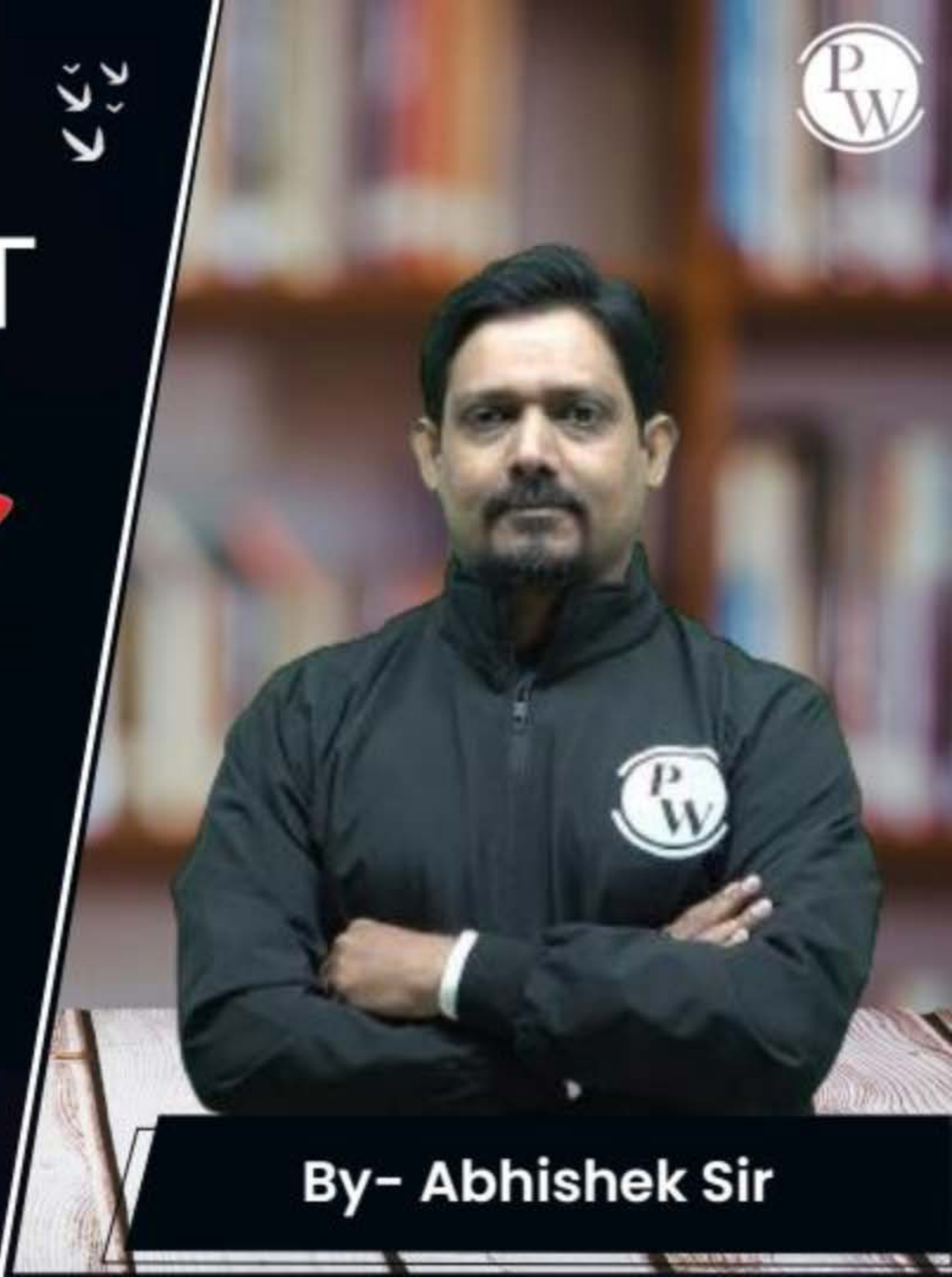
Computer Science & IT



Fundamental of C Language

String & Structure

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



Topic — 2-Dimensional array

Topic — Rows & columns

Topic — Arrangment of elements

Topic — printing value of 1-D & 2-D array

Topic

Slide

Topics to be Covered



- Topic — string
- Topic — string declaration ✓
- Topic — string print ✓
- Topic — stolen function ✓
- Topic

Slide



String

1. String is a Token : Any thing between double quotes (" ")

2. string is array of characters :

3. The end of string is a special character is called as NULL character.

\0 - NULL character

ASCII value is zero

Slide



String

```
char ch = '\0';
printf("%c", ch); // No output
printf("%d", ch); // 0 - NULL character ASCII value
```

At the end of NULL string NULL character is present



String

string declaration & initialization

```
char str[6] = "vijay";
```

v	i	j	a	y	\0

It work as boundary for strng

Slide



String

```
char ch[6] = {'v','i','j','a','y','\0'};
```

we can declare a string using character pointer.

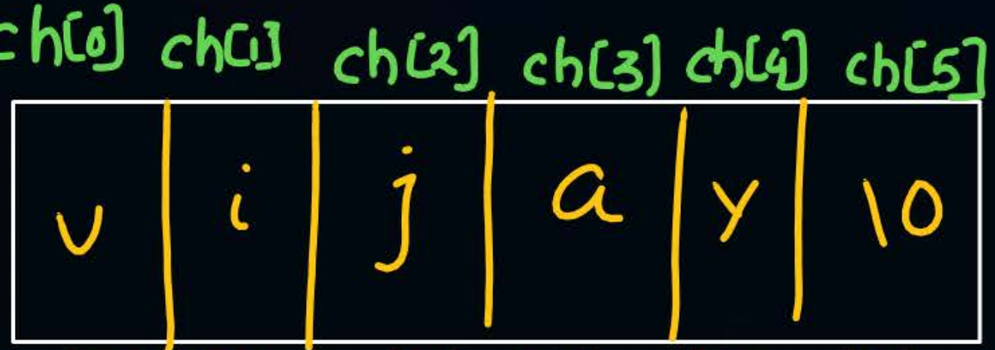
Address of a character.

Slide

String

String is array of character

```
char ch[6] = "vijay";
```




```
printf("%c", ch[1]); // i printed
printf("%d", ch[1]); // ASCII value printed
                        105
printf("%c", ch[3]); // a printed.
```

a - 97
b - 98
c - 99
d - 100
e - 101
f - 102

g - 103
h - 104
i - 105

Slide



String

Character printed without any quotes. a - output

To print the string format specifier %s is used and we Need to provide Address of first character

Slide



String

printf ("%s", &ch[0]); & Address of operator

prints everything till it gets the NULL character

vijay

printf ("%s", &ch[2]); // jay will be printed

Slide



String

String related function are defined in header file string.h

```
#include<stdio.h>
#include<string.h>

int main(){

  char ch[6] = "vijay";

  printf("%c", ch[1]);
  printf("%s", &ch[0]);

  printf("%lu", strlen(&ch[0]));

  return 0;
}
```



String

lu :- Long unsigned

```
char ch[6] = "vijay";

printf("%lu", strlen(&ch[0]));
```

Strlen

* string length
* if length required then give address of first character
* it return unsigned value representing length of string.

Slide



2 mins Summary

Topic — String as array of character

Topic — '\0' - NULL character

Topic — String declaration & initialization

Topic — String print

Topic — strlen() function, string.h

Slide

THANK - YOU

